



KWAZULU-NATAL PROVINCE

EDUCATION

REPUBLIC OF SOUTH AFRICA

**NATIONAL
SENIOR CERTIFICATE**

GRADE 11

MATHEMATICS P2

NOVEMBER 2025

MARKS: 150

TIME: 3 hours

This question paper consists of 13 pages and 1 information sheet.

INSTRUCTIONS AND INFORMATION

Read the following instructions carefully before answering the questions.

1. This question paper consists of 11 questions.
2. Answer ALL the questions in the SPECIAL ANSWER BOOK provided.
3. Clearly show numbering if additional space has been used to answer a question. If NOT, cross out all rough work done in the additional space.
4. Clearly show ALL calculations, diagrams, graphs, etc. which you have used in determining your answers.
5. Answers only will NOT necessarily be awarded full marks.
6. You may use an approved scientific calculator (non-programmable and non-graphical), unless stated otherwise.
7. If necessary, round off answers correct to TWO decimal places, unless stated otherwise.
8. Diagrams are NOT necessarily drawn to scale.
9. An information sheet with formulae is included at the end of the question paper.
10. Write neatly and legibly.

QUESTION 1

Learners collected cans during "Operation Clean Up". The number of cans collected by 10 classes is shown in the table below.

250	259	265	276	290	290	294	310	315	330
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- 1.1 Calculate the mean number of cans collected by these 10 classes. (2)
- 1.2 Write down the standard deviation of the above data. (1)
- 1.3 How many classes are within one standard deviation of the mean? (2)
- 1.4 Draw a box-and-whisker diagram of the data on the scaled line provided in the ANSWER BOOK. (2)
- 1.5 When another 10 classes added their cans to the original number of cans collected, the new mean is 250.
 - 1.5.1 Calculate how many cans were collected by the 10 classes that were added. (2)
 - 1.5.2 What effect does adding the cans of these 10 classes have on the original standard deviation? (1)

[10]

QUESTION 2

Learners were asked the time (to the nearest minute) that they usually take to get from home to school each morning. The results are shown in the table below:

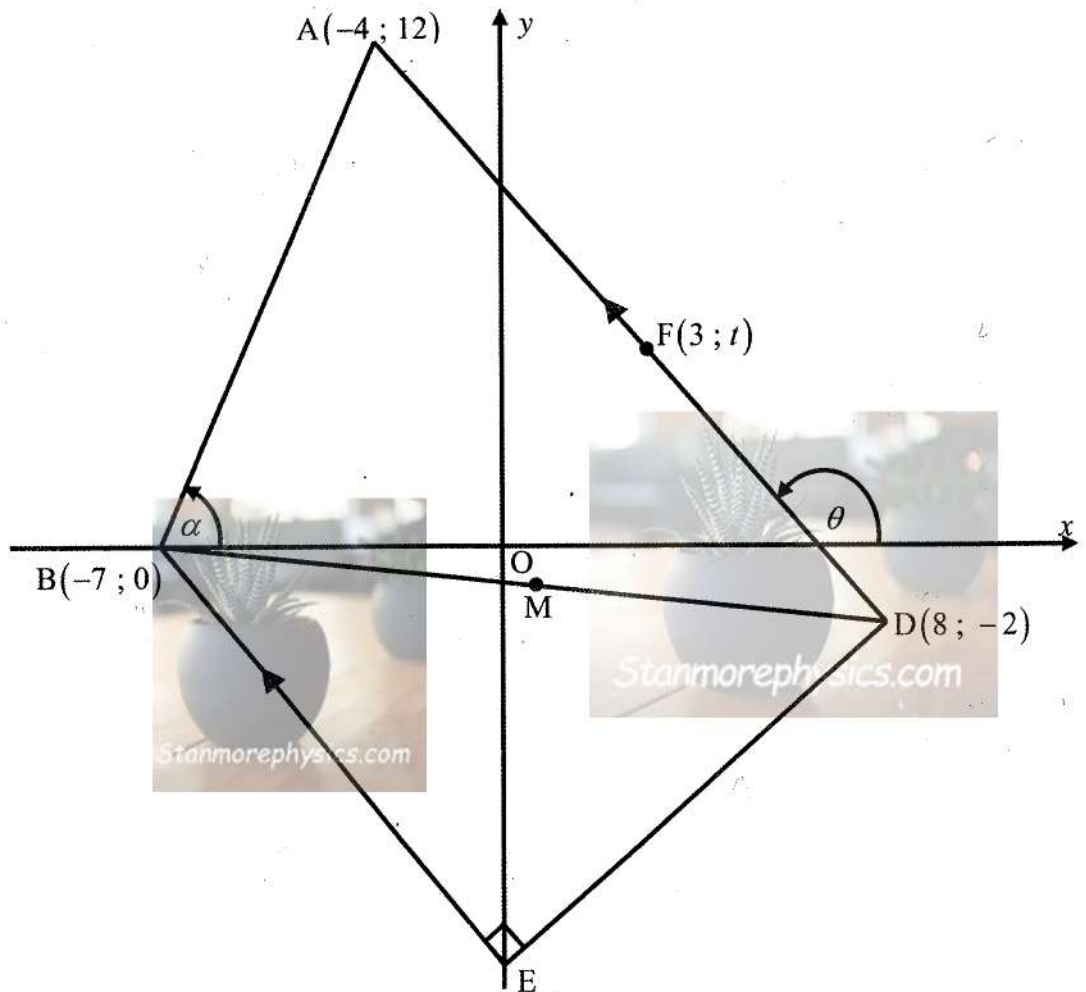
Time in minutes	Number of learners
$5 < t \leq 10$	160
$10 < t \leq 15$	150
$15 < t \leq 20$	110
$20 < t \leq 25$	60
$25 < t \leq 30$	45
$30 < t \leq 35$	15

- 2.1 Write down the modal class of the data. (1)
- 2.2 Complete the cumulative frequency column in the table provided in the ANSWER BOOK. (1)
- 2.3 Draw the ogive of this data on the grid provided in the ANSWER BOOK. (3)
- 2.4 Estimate the percentage of learners who take more than 25 minutes to get to school. (2)
- 2.5 Estimate the mean time that these learners take to get to school in the morning. (3)

[10]

QUESTION 3

In the diagram below, $A(-4; 12)$, $B(-7; 0)$, E and $D(8; -2)$ are the vertices of a trapezium. E lies on the y axis such that BE is perpendicular to DE . The equation of AB is $y = 4x + 28$. The angles of inclination of AB and AD are α and θ respectively. M is the midpoint of BD . $F(3; t)$ is a point on AD .

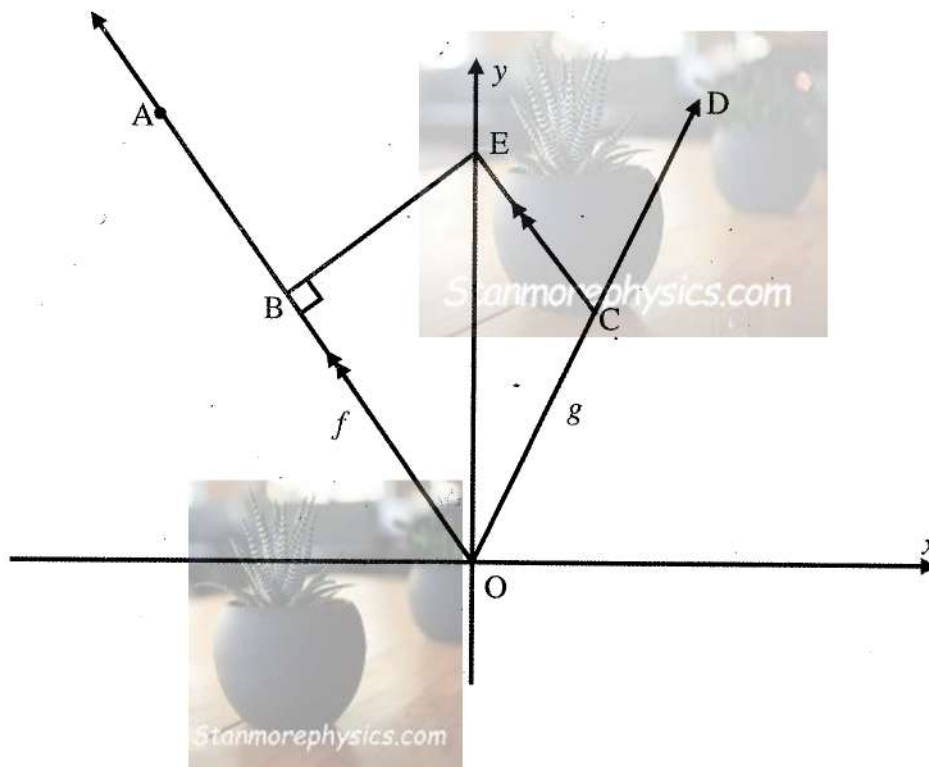


- 3.1 Calculate the length of AD. (2)
- 3.2 Calculate the coordinates of M, the midpoint of BD. (2)
- 3.3 Calculate the gradient of BE. (3)
- 3.4 Determine the equation of DE. (3)
- 3.5 Calculate the value of t , the y -coordinate of F. (3)
- 3.6 Calculate the coordinates of C, if ABCD, in that order, is a parallelogram. (2)
- 3.7 Calculate the size of $\hat{BCD}$, correct to the nearest degree. (5)

[20]

QUESTION 4

In the diagram below, A is a point on the line with equation $y + 2x = 0$. B is a point on OA such that $OB = 5\sqrt{5}$ units. E is a point on the y axis such that EB is perpendicular to OA. The equation of OD is $y = 2x$ and $CE \parallel OA$.

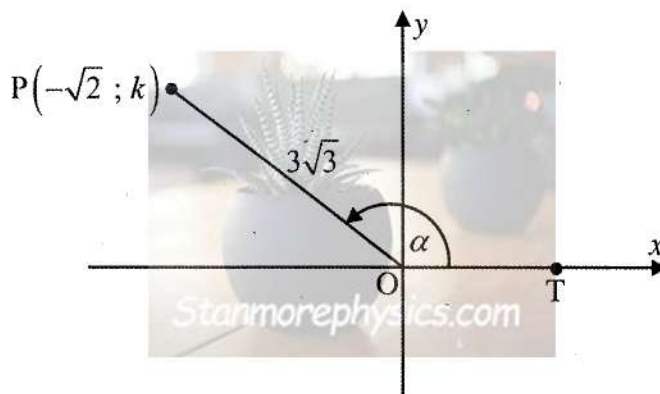


Determine the following:

- 4.1 The coordinates of B. (4)
 - 4.2 The area of the trapezium OBEC correct to the nearest square unit. (8)
- [12]

QUESTION 5

- 5.1 In the diagram below, P is the point $(-\sqrt{2}; k)$ such that $OP = 3\sqrt{3}$ units. $\hat{TOP} = \alpha$ is an obtuse angle.



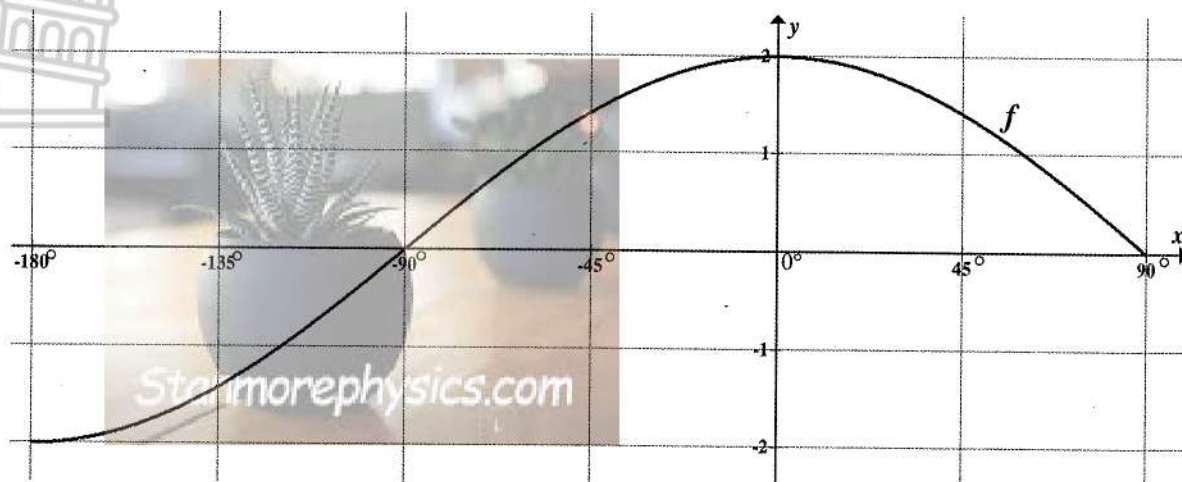
WITHOUT using a calculator, determine the value of the following:

- 5.1.1 k . (2)
- 5.1.2 $\tan(\alpha - 180^\circ)$ (2)
- 5.1.3 $\cos \beta$, if it is further given that $\alpha + \beta = 180^\circ$. (2)
- 5.2 If $\sin 34^\circ = p$, determine, in terms of p , the value of $\tan 484^\circ$ **WITHOUT using a calculator**. (4)
- 5.3 Simplify the following **WITHOUT the use of a calculator**: (5)
- $$\frac{\tan 205^\circ \cdot \cos 315^\circ \cdot \sin(-45^\circ)}{\sin 210^\circ \cdot \cos 150^\circ \cdot \tan 25^\circ}$$
- 5.4 Simplify the following to a single trigonometric term: (6)
- $$\sin(180^\circ + x) \cdot \cos(x - 90^\circ) - \frac{\sin x}{\sin(90^\circ - x) \cdot \tan(180^\circ - x)}$$
- 5.5 Given the identity: $\frac{1 + \sin x}{\cos x} + \frac{\cos x}{1 + \sin x} = \frac{2}{\cos x}$
- 5.5.1 Prove the identity. (4)
- 5.5.2 For which values of x in the interval $0^\circ \leq x \leq 360^\circ$ will the above identity be undefined? (2)
- 5.6 Determine the general solution of the equation: $\tan x \cdot \sin x = \frac{3}{\sin x} - \cos x$ (6)

[33]

QUESTION 6

Below is a sketch of the graph of $f(x) = a \cos x$ for $x \in [-180^\circ; 90^\circ]$.

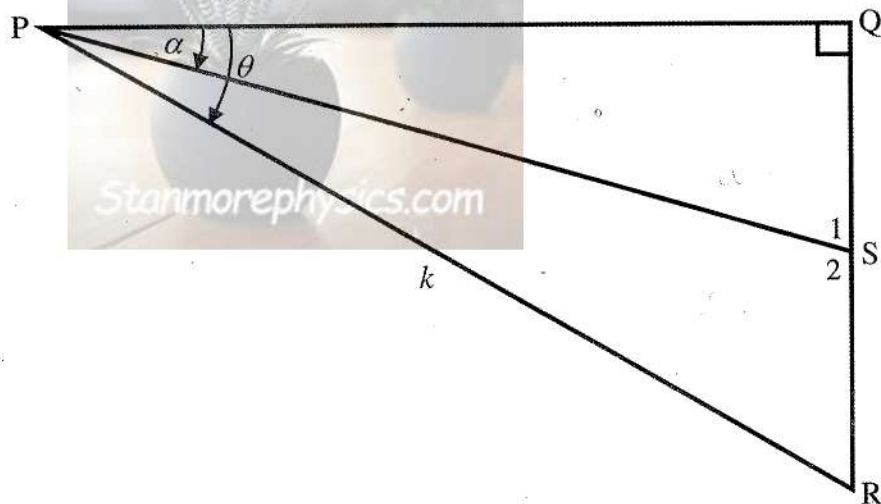


- 6.1 Write down the value of a . (1)
- 6.2 Write down the period of $g(x) = -\cos 2x$. (1)
- 6.3 Sketch the graph of $g(x) = -\cos 2x$ for $x \in [-180^\circ; 90^\circ]$ on the set of axes provided in the ANSWER BOOK. (3)
- 6.4 Use the graphs to write down the value of x for which $f(x) - 1 = g(x) + 2$. (2)
- 6.5 Determine minimum value of $1 - 2[\sin(90^\circ - x)]$. (3)

[10]

QUESTION 7

In the diagram below, $PQ \perp QR$. $\hat{QPS} = \alpha$ and $\hat{QPR} = \theta$ and $PR = k$ units.



7.1 Write down the size of $\hat{RPS}$ in terms of α and θ .

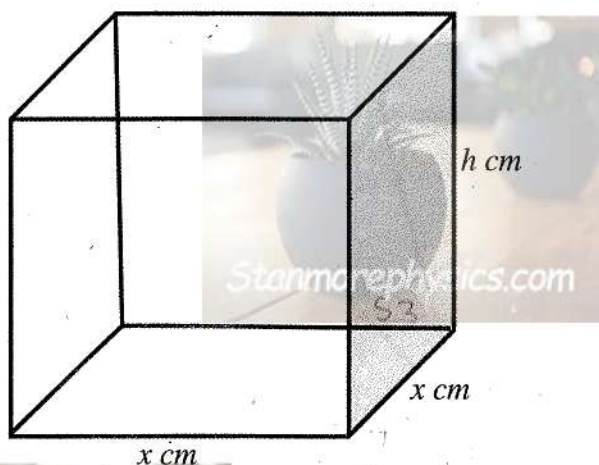
7.2 Show that $RS = \frac{k \cdot \sin(\theta - \alpha)}{\cos \alpha}$

7.3 Calculate the length of RS if $k = 17$ units, $\theta = 58^\circ$ and $\alpha = 23^\circ$.

7.4 Show that the area of $\triangle RSP = 48$ square units, correct to the nearest square unit.

QUESTION 8

The diagram below represents an open tank with a square base (side dimensions of $x \text{ cm}$) and a height of $h \text{ cm}$. The tank has a volume of 490 cm^3 .

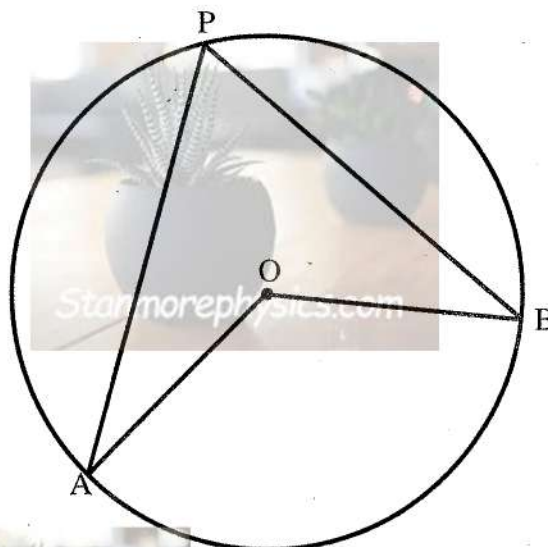


- 8.1 Determine the height, h , of the tank in terms of x . (2)
- 8.2 Show that A , the external surface area of the tank is given by: $A = x^2 + \frac{1960}{x} \text{ cm}^2$. (2)
- 8.3 Calculate the external surface area of the tank if the height is 10 cm . (4)

[8]

QUESTION 9

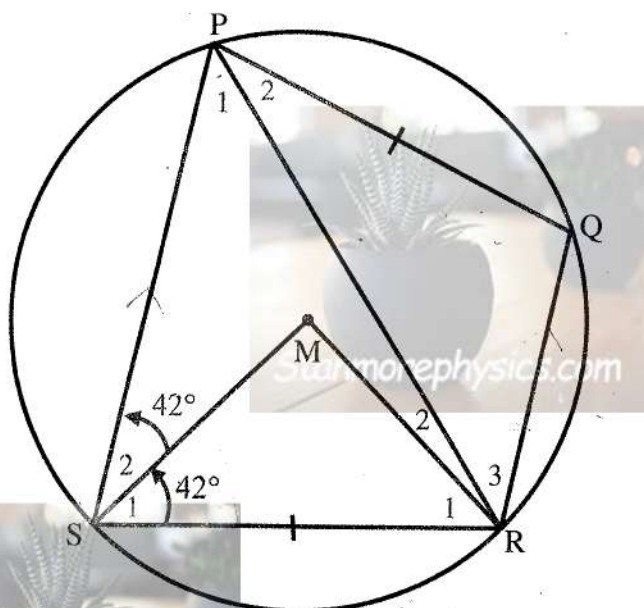
- 9.1 In the diagram below, A, B and P are points on the circle with centre O. PA, PB, AO and BO are joined.



Use the diagram in the ANSWER BOOK to prove the theorem which states that the angle subtended by an arc or chord at the centre of the circle is twice the angle subtended by the same arc or chord at the circumference, that is, prove that $\hat{AOB} = 2\hat{APB}$.

9.2

In the diagram below, PQRS is a cyclic quadrilateral. M is the centre of the circle. MS bisects $\hat{PSR}$ such that $\hat{S}_1 = \hat{S}_2 = 42^\circ$ and $PQ = SR$.



9.2.1 Name, giving a reason, another angle equal to 42° . (2)

9.2.2 Calculate, giving reasons, the sizes of:

(a) $\hat{PQR}$ (2)

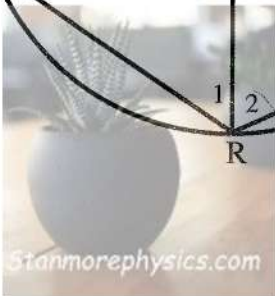
(b) $\hat{P}_1$ (3)

9.2.3 Prove that $SP \parallel QR$. (3)

[15]

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chord TZ. $\hat{T}_3 = x$.

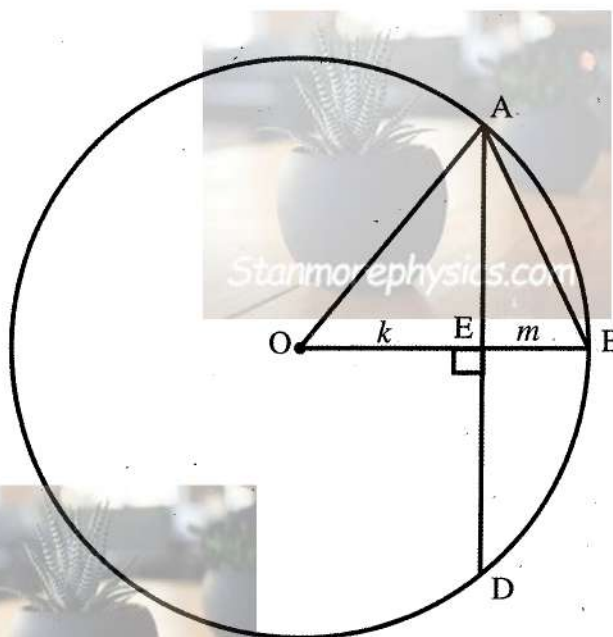


Prove that:

- 10.1 TOZV is a cyclic quadrilateral.
- 10.2 $RZ \parallel SY$
- 10.3 $\hat{T}_1 = \frac{1}{2}x$
- 10.4 $\hat{Y}_2 = 90^\circ + \hat{T}_1$

QUESTION 11

In the diagram below, AD is a chord of circle with centre O. OB is drawn perpendicular to AD and cuts AD at E. $OE = k$ and $EB = m$. OA and AB are drawn.



11.1 Prove that $AB^2 = 2km + 2m^2$. (4)

11.2 Write down, giving a reason, the length of DE in terms of k and m . (2)

[6]

TOTAL: 150 MARKS

INFORMATION SHEET

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$A = P(1 + ni)$$

$$A = P(1 - ni)$$

$$A = P(1 - i)^n$$

$$A = P(1 + i)^n$$

$$T_n = a + (n - 1)d$$

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}; r \neq 1$$

$$S_\infty = \frac{a}{1 - r}; -1 < r < 1$$

$$F = \frac{x[(1 + i)^n - 1]}{i}$$

$$P = \frac{x[1 - (1 + i)^{-n}]}{i}$$

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$$

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$M\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

$$y = mx + c$$

$$y - y_1 = m(x - x_1)$$

$$m = \frac{y_2 - y_1}{x_2 - x_1}$$

$$m = \tan \theta$$

$$(x - a)^2 + (y - b)^2 = r^2$$

$$\text{In } \triangle ABC: \frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

$$a^2 = b^2 + c^2 - 2bc \cos A$$

$$\text{area } \triangle ABC = \frac{1}{2} ab \sin C$$

$$\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$$

$$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$$

$$\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$$

$$\cos(\alpha - \beta) = \cos \alpha \cos \beta + \sin \alpha \sin \beta$$

$$\cos 2\alpha = \begin{cases} \cos^2 \alpha - \sin^2 \alpha \\ 1 - 2\sin^2 \alpha \\ 2\cos^2 \alpha - 1 \end{cases}$$

$$\sin 2\alpha = 2\sin \alpha \cos \alpha$$

$$\bar{x} = \frac{\sum_{i=1}^n x_i}{n}$$

$$\sigma^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n}$$

$$P(A) = \frac{n(A)}{n(S)}$$

$$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$$

$$\hat{y} = a + bx$$

$$b = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2}$$



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**NATIONAL
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GRADE 11

MATHEMATICS P2

NOVEMBER 2025

Stanmorephysics.com

MARKING GUIDELINES

Stanmorephysics.com

MARKS: 150

These marking guidelines consist of 18 pages.

NOTE:

GRADE 11

Marking Guideline

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- Consistent accuracy applies in ALL aspects of the Marking Guidelines. Stop marking at the second calculation error.
- Assuming answers/values in order to solve a problem is NOT acceptable.

NOTA:

- *As 'n kandidaat 'n vraag TWEE KEER beantwoord, merk slegs die EERSTE poging.*
- *Volgehoue akkuraatheid word in ALLE aspekte van die Nasienriglyne toegepas. Hou op nasien by die tweede berekeningsfout.*
- *Aanvaar van antwoorde/waardes om 'n probleem op te los, word NIE toegelaat nie.*

GEOMETRY	
S	A mark for a correct statement (A statement mark is independent of a reason)
	'n Punt vir 'n korrekte bewering ('n Punt vir 'n bewering is onafhanklik van die rede)
R	A mark for the correct reason (A reason mark may only be awarded if the statement is correct)
	'n Punt vir 'n korrekte rede ('n Punt word slegs vir die rede toegeken as die bewering korrek is)
S/R	Award a mark if statement AND reason are both correct
	Ken 'n punt toe as die bewering EN rede beide korrek is

QUESTION 1

1.1	$\bar{x} = \frac{2879}{10}$ $= 287,9$	A✓ 2879 CA✓ answer (2)
1.2	$\sigma = 24,45$	A✓ $\sigma = 24,45$ (1)
1.3	Interval: (263,45 ; 312,35) $\therefore$ 6 classes	CA✓ $(\bar{x} - \sigma ; \bar{x} + \sigma)$ CA✓ answer (2)
1.4		A✓ whiskers ending at 250 & 330 A✓ box with $Q_1=265$ $Q_2=290$ & $Q_3=310$ (2)
1.5.1	Let the number of cans added be k . $\therefore \frac{2879+k}{20} = 250$ $2879+k = 5000$ $k = 2121$ $\therefore$ 2121 cans were added.	CA✓ $\frac{2879+k}{20} = 250$ CA✓ answer (2)
1.5.2	The standard deviation will increase.	A✓ increase (1)
		[10]

QUESTION 2

2.1	$5 < t \leq 10$	A✓ answer (1)																					
2.2	<table border="1"> <thead> <tr> <th>Time (in minutes)</th><th>Number of learners</th><th>Cumulative frequency</th></tr> </thead> <tbody> <tr> <td>$5 < t \leq 10$</td><td>160</td><td>160</td></tr> <tr> <td>$10 < t \leq 15$</td><td>150</td><td>310</td></tr> <tr> <td>$15 < t \leq 20$</td><td>110</td><td>420</td></tr> <tr> <td>$20 < t \leq 25$</td><td>60</td><td>480</td></tr> <tr> <td>$25 < t \leq 30$</td><td>45</td><td>525</td></tr> <tr> <td>$30 < t \leq 35$</td><td>15</td><td>540</td></tr> </tbody> </table>	Time (in minutes)	Number of learners	Cumulative frequency	$5 < t \leq 10$	160	160	$10 < t \leq 15$	150	310	$15 < t \leq 20$	110	420	$20 < t \leq 25$	60	480	$25 < t \leq 30$	45	525	$30 < t \leq 35$	15	540	A✓ all cumulative frequency values correct (1)
Time (in minutes)	Number of learners	Cumulative frequency																					
$5 < t \leq 10$	160	160																					
$10 < t \leq 15$	150	310																					
$15 < t \leq 20$	110	420																					
$20 < t \leq 25$	60	480																					
$25 < t \leq 30$	45	525																					
$30 < t \leq 35$	15	540																					
2.3		A✓ grounding at (5;0) CA✓ plotting cumulative frequency at upper limits A✓ smooth increasing curve (3)																					
2.4	No. of learners = $540 - 480$ = 60 learners % of learners = $\frac{60}{540} \times 100\%$ = 11,11%	A✓ 60 CA✓ answer (2)																					
2.5	$\bar{x} = \frac{(7,5 \times 160) + (12,5 \times 150) + (17,5 \times 110) + (22,5 \times 60) + (27,5 \times 45) + (32,5 \times 15)}{540}$ $= \frac{8075}{540}$ = 14,95 minutes ANSWER ONLY: Full marks. All ACCURACY marks Answer must EXACTLY be 14,95 minutes.	A✓ frequency × midpoints A✓ 8075 CA✓ answer (3)																					
		[10]																					

QUESTION 3

3.1.	$AD = \sqrt{[-4-8]^2 + [12-(-2)]^2}$ $= \sqrt{340} \text{ or } 2\sqrt{85} \text{ units or } 18,44 \text{ units}$	A✓ substitution in correct formula CA✓ answer (2)
3.2	$M\left(\frac{-7+8}{2}; \frac{0-2}{2}\right)$ $M\left(\frac{1}{2}; -1\right)$	A✓ x-coordinate of $\frac{1}{2}$ A✓ y-coordinate of -1 (2)
3.3	$m_{BE} = m_{AD} \quad (BE \parallel AD)$ $= \frac{12-(-2)}{-4-8}$ $= -\frac{14}{12} \text{ or } -\frac{7}{6}$ If $m_{BE} = m_{AD}$ is not stated: Award 2 marks here	A✓ $m_{BE} = m_{AD}$ A✓ substitution CA✓ answer (3)
3.4	$m_{DE} = \frac{6}{7} (BE \perp DE)$ $y - y_1 = m(x - x_1)$ $y + 2 = \frac{6}{7}(x - 8)$ $y + 2 = \frac{6}{7}x - \frac{48}{7}$ $y = \frac{6}{7}x - \frac{62}{7}$ OR $m_{DE} = \frac{6}{7} (DE \perp DE)$ $y = mx + c$ $-2 = \frac{6}{7}(8) + c$ $c = -\frac{62}{7}$ $y = \frac{6}{7}x - \frac{62}{7}$ If $m_{DE} = -\frac{1}{m_{BE}}$ is not stated: Award 2 marks here	CA✓ $m_{DE} = -\frac{1}{m_{BE}}$ CA✓ substitution of $D(8; -2)$ & m_{DE} into equation of a line CA✓ answer (3)
	If $m_{DE} = -\frac{1}{m_{BE}}$ is not stated: Award 2 marks here	CA✓ substitution of $D(8; -2)$ & m_{DE} into equation of a line CA✓ answer (3)
3.5	$m_{AF} = m_{FD}$ $\frac{t-12}{3-(-4)} = \frac{t+2}{3-8}$ $7t+14 = -5t+60$ $12t = 46$ $t = \frac{23}{6} \text{ or } 3\frac{5}{6} \text{ or } 3,83$	A✓ condition A✓ substitution CA✓ answer (3)

GRADE 11
Marking Guideline

	OR $m_{AD} = -\frac{7}{6}$ $\frac{t - (-2)}{3 - 8} = -\frac{7}{6}$ $t + 2 = -5\left(-\frac{7}{6}\right)$ $t = \frac{23}{6} \text{ or } 3\frac{5}{6} \text{ or } 3,83$	A✓ condition A✓ substitution CA✓ answer (3)
3.5	OR Equation of AD: $y - 12 = -\frac{7}{6}[x - (-4)] \text{ or } 12 = -\frac{7}{6}(-4) + c$ $y - 12 = -\frac{7}{6}x - \frac{28}{6} \text{ or } c = \frac{22}{3}$ $y = -\frac{7}{6}x + \frac{22}{3}$ $t = -\frac{7}{6}(3) + \frac{22}{3}$ $t = \frac{23}{6} \text{ or } 3\frac{5}{6} \text{ or } 3,83$ OR Equation of AD: $y - (-2) = -\frac{7}{6}(x - 8) \text{ or } -2 = -\frac{7}{6}(8) + c$ $y + 2 = -\frac{7}{6}x + \frac{28}{3} \text{ or } c = \frac{22}{3}$ $y = -\frac{7}{6}x + \frac{22}{3}$ $t = -\frac{7}{6}(3) + \frac{22}{3}$ $t = \frac{23}{6} \text{ or } 3\frac{5}{6} \text{ or } 3,83$	A✓ substitution of A(-4 ; 12) & m_{AD} into equation of a line A✓ substitution of F(3 ; t) CA✓ answer (3) A✓ substitution of D(8 ; -2) & m_{AD} into equation of a line A✓ substitution of F(3 ; t) CA✓ answer (3)
3.6	C(5 ; -14) [by translation] OR $C(x ; y) \quad M\left(\frac{1}{2} ; -1\right) \quad A(-4 ; 12)$ $\frac{x_C - 4}{2} = \frac{1}{2} \text{ and } \frac{y_C + 12}{2} = -1$ $x_C - 4 = 1 \text{ and } y_C + 12 = -2$ $x_C = 5 \text{ and } y_C = -14$ $\therefore C(5 ; -14)$	AA✓✓ answer (2) AA✓✓ answer (2)

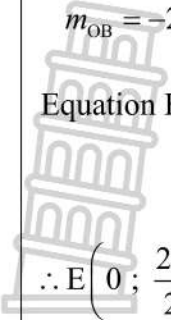
GRADE 11
Marking Guideline

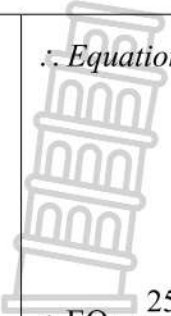
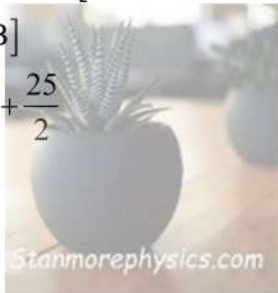
3.7	$\tan \theta = m_{AD}$ $= -\frac{7}{6}$ $\theta = 130,60^\circ$ $\tan \alpha = m_{AB} = 4$ $\alpha = 75,96^\circ$ $\hat{B}AD = 130,60^\circ - 75,96^\circ$ [ext. $\angle$ of Δ] $= 54,64^\circ$ $\therefore \hat{B}CD = \hat{B}AD = 55^\circ$ [opp. $\angle$ s of a parm.]	CA✓ $\tan \theta = m_{AD}$ CA✓ θ A✓ α CA✓ $\hat{B}AD$ CA✓ answer (5)
		[20]

QUESTION 4

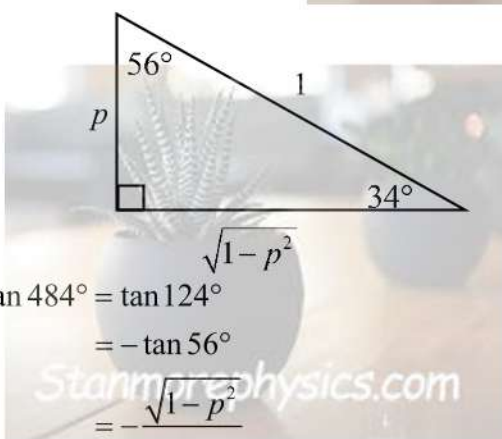
4.1	Equation of AO: $y = -2x \therefore B(x; -2x)$ $OB = 5\sqrt{5}$ $\sqrt{(x-0)^2 + (-2x-0)^2} = 5\sqrt{5}$ $\sqrt{x^2 + 4x^2} = 5\sqrt{5}$ $5x^2 = 125$ $x^2 = 25$ $x = -5$ $\therefore B(-5; 10)$	A✓ substitution of $O(0; 0)$ & $B(x; -2x)$ into distance formula A✓ equating to $5\sqrt{5}$ CA✓ x value CA✓ y value (4)
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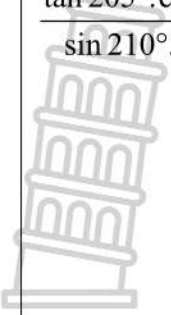
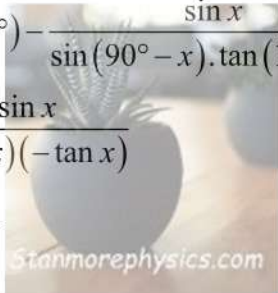
GRADE 11
Marking Guideline

4.2	 $m_{OB} = -2 \quad \therefore m_{BE} = \frac{1}{2} \quad (EB \perp OA)$ Equation BE: $y - 10 = \frac{1}{2}(x + 5)$ $\therefore y = \frac{1}{2}x + \frac{25}{2}$ $\therefore E\left(0; \frac{25}{2}\right)$ $m_{EC} = -2 \quad [EC \parallel OB]$ Equation of EC: $y = -2x + \frac{25}{2}$ at C: $-2x + \frac{25}{2} = 2x$ $x = \frac{25}{8}$ $y = 2\left(\frac{25}{8}\right) = \frac{25}{4}$ $\therefore C\left(\frac{25}{8}; \frac{25}{4}\right)$  $EC = \sqrt{\left(\frac{25}{8} - 0\right)^2 + \left(\frac{25}{4} - \frac{25}{2}\right)^2} = \frac{25\sqrt{5}}{8} \text{ units}$ $BE = \sqrt{(-5 - 0)^2 + \left(10 - \frac{25}{2}\right)^2} = \frac{5\sqrt{5}}{2}$ Area OBEC = $\frac{1}{2}(BO + EC)(BE)$ $= \frac{1}{2}\left(5\sqrt{5} + \frac{25\sqrt{5}}{8}\right)\left(\frac{5\sqrt{5}}{2}\right)$ $= 51 \text{ square units}$	A✓ $m_{BE} = \frac{1}{2}$ CA✓ $E\left(0; \frac{25}{2}\right)$ CA✓ equation of EC CA✓ $C\left(\frac{25}{8}; \frac{25}{4}\right)$ CA✓ length of EC CA✓ length of BE CA✓ substitution into area of a trapezium CA✓ answer (8)
	OR $\tan \theta = m_{OB} = -2$ $\therefore \theta = 180^\circ - \tan^{-1}(2)$ $\theta = 116,57^\circ$ $\therefore \hat{BOE} = 116,57 - 90^\circ$ $= 26,57^\circ$ $m_{BE} = \frac{1}{2}$	A✓ $\theta = 116,57^\circ$ CA✓ size of $\hat{BOE}$

 $\therefore \text{Equation BE: } y - 10 = \frac{1}{2}(x + 5)$ $\therefore y = \frac{1}{2}x + \frac{25}{2}$ $\therefore E\left(0; \frac{25}{2}\right)$ $\therefore EO = \frac{25}{2} \text{ units and } OB = 5\sqrt{5} \text{ units}$ $\text{Area } \triangle BOE = \frac{1}{2} \times \left(\frac{25}{2}\right) \times (5\sqrt{5}) \times \sin 26,57^\circ$ $= 31,26 \text{ square units}$ In $\triangle EOC$: $\hat{CEO} = 26,57^\circ$ [alt $\angle$s; $BO \parallel EC$] $m_{EC} = -2 \quad [EC \parallel OB]$ Equation of EC: $y = -2x + \frac{25}{2}$ at E: $2x = -2x + \frac{25}{2}$ $x = \frac{25}{8}$ $y = 2\left(\frac{25}{8}\right)$ $y = \frac{25}{4}$ $\therefore C\left(\frac{25}{8}; \frac{25}{4}\right)$ $EC = \sqrt{\left(\frac{25}{8} - 0\right)^2 + \left(\frac{25}{4} - \frac{25}{2}\right)^2} = \frac{25\sqrt{5}}{8}$ $\text{Area } \triangle EOC = \frac{1}{2} \times \left(\frac{25}{2}\right) \times \left(\frac{25\sqrt{5}}{8}\right) \times \sin 26,57^\circ$ $= 19,53 \text{ square units}$ OR $\text{Area } \triangle EOC = \frac{1}{2} \times \left(\frac{25}{2}\right) \times \left(\frac{25}{8}\right)$ $= 19,53 \text{ square units}$ $\therefore \text{Area OBEC} = 31,26 + 19,53$ $= 51 \text{ square units}$  	CA✓ coordinates of E CA✓ substitution into area formula CA✓ Area $\triangle BOE$ CA✓ coordinates of C CA✓ Area $\triangle EOC$ OR CA✓ Area $\triangle EOC$ CA✓ answer (8)
	[12]

QUESTION 5

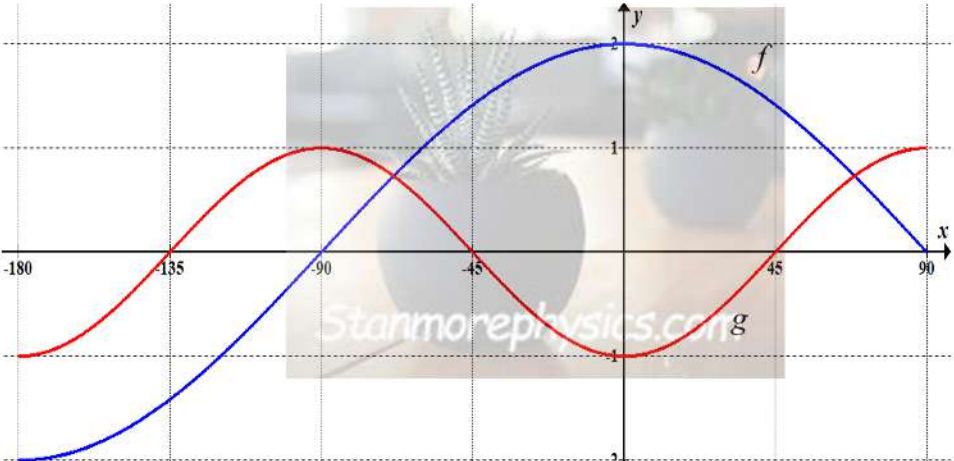
5.1.1	$\begin{aligned} (-\sqrt{2})^2 + k^2 &= (3\sqrt{3})^2 \\ 2 + k^2 &= 27 \\ k^2 &= 25 \\ k &= \pm 5 \\ \therefore k &= 5 \end{aligned}$	A✓ substitution CA✓ answer (2)
5.1.2	$\begin{aligned} \tan(\alpha - 180^\circ) &= \tan \alpha \\ &= -\frac{5}{\sqrt{2}} \end{aligned}$	A✓ $\tan \theta$ CA✓ answer (2)
5.1.3	$\begin{aligned} \cos \beta &= \cos(180^\circ - \alpha) \\ &= -\cos \alpha \\ &= -\frac{(-\sqrt{2})}{3\sqrt{3}} \\ &= \frac{\sqrt{2}}{3\sqrt{3}} \end{aligned}$ 	A✓ $-\cos \alpha$ A✓ answer (2)
5.2	$\sin 34^\circ = p$  $\tan 484^\circ = \tan 124^\circ$ $= -\tan 56^\circ$ $= -\frac{\sqrt{1-p^2}}{p}$ OR $\tan 484^\circ = \tan 124^\circ$ $= \frac{\sin 124^\circ}{\cos 124^\circ}$ $= \frac{\sin(90^\circ + 34^\circ)}{\cos(90^\circ + 34^\circ)}$ $= \frac{\cos 34^\circ}{-\sin 34^\circ}$ $= -\frac{\sqrt{1-p^2}}{p}$	A✓ third side = $\sqrt{1-p^2}$ A✓ $\tan 124^\circ$ CA✓ reduction CA✓ answer OR A✓ $\tan 124^\circ$ A✓ $\frac{\sin 124^\circ}{\cos 124^\circ}$ CA✓ $\frac{\cos 34^\circ}{-\sin 34^\circ}$ CA✓ answer (4)

5.3	 $\frac{\tan 205^\circ \cdot \cos 315^\circ \cdot \sin(-45^\circ)}{\sin 210^\circ \cdot \cos 150^\circ \cdot \tan 25^\circ} = \frac{(\tan 25^\circ)(\cos 45^\circ)(-\sin 45^\circ)}{(-\sin 30^\circ)(-\cos 30^\circ)(\tan 25^\circ)}$ $= \left(\frac{\sqrt{2}}{2}\right)\left(-\frac{\sqrt{2}}{2}\right)$ $= \left(-\frac{1}{2}\right)\left(-\frac{\sqrt{3}}{2}\right)$ $= -\frac{1}{2}$ $= -\frac{2}{\sqrt{3}}$ $= -\frac{2}{\sqrt{3}}$	A✓ reductions in numerator A✓ reductions in denominator CA✓ substitution of special $\angle$ values for 45° CA✓ substitution of special $\angle$ values for 30° CA✓ answer (5)
5.4	 $\sin(180^\circ + x) \cdot \cos(x - 90^\circ) - \frac{\sin x}{\sin(90^\circ - x) \cdot \tan(180^\circ - x)}$ $= (-\sin x)(\sin x) - \frac{\sin x}{(\cos x)(-\tan x)}$ $= -\sin^2 x + \frac{\sin x}{\cos x \left(\frac{\sin x}{\cos x}\right)}$ $= -\sin^2 x + 1$ $= \cos^2 x$	A✓ $\sin(180^\circ + x) = -\sin x$ A✓ $\cos(x - 90^\circ) = \sin x$ A✓ $\sin(90^\circ - x) = \cos x$ A✓ $\tan(180^\circ - x) = -\tan x$ A✓ $\tan x = \frac{\sin x}{\cos x}$ CA✓ answer (6)
5.5.1	$\text{LHS} = \frac{1 + \sin x}{\cos x} + \frac{\cos x}{1 + \sin x}$ $= \frac{(1 + \sin x)^2 + (\cos x)^2}{\cos x(1 + \sin x)}$ $= \frac{1 + 2\sin x + \sin^2 x + \cos^2 x}{\cos x(1 + \sin x)}$ $= \frac{1 + 2\sin x + 1}{\cos x(1 + \sin x)}$ $= \frac{2 + 2\sin x}{\cos x(1 + \sin x)}$ $= \frac{2(1 + \sin x)}{\cos x(1 + \sin x)}$ $= \frac{2}{\cos x}$ $= \text{RHS}$	A✓ numerator A✓ LCD A✓ identity: $\sin^2 x + \cos^2 x = 1$ A✓ factorization (4)
5.5.2	Undefined: $\cos x = 0$ or $\sin x = -1$ $x = 90^\circ$ or $x = 270^\circ$ or $x = 270^\circ$ $\therefore x \in \{90^\circ ; 270^\circ\}$	✓A 90° ✓A 270° (2)

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5.6	$\tan x \cdot \sin x = \frac{3}{\sin x} - \cos x$ $\frac{\sin x}{\cos x} \cdot \sin x = \frac{3}{\sin x} - \cos x$ $\sin^3 x = 3 \cos x - \sin x \cdot \cos^2 x$ $\sin^3 x + \sin x \cdot \cos^2 x = 3 \cos x$ $\sin x (\sin^2 x + \cos^2 x) = 3 \cos x$ $\frac{\sin x(1)}{\cos x} = 3$ $\tan x = 3$ $x = 71,57^\circ + k \cdot 180; k \in \mathbb{Z}$ OR $\tan x \cdot \sin x = \frac{3}{\sin x} - \cos x$ $\frac{\sin x}{\cos x} \cdot \frac{\sin x}{1} + \cos x = \frac{3}{\sin x}$ $\frac{\sin^2 x + \cos^2 x}{\cos x} = \frac{3}{\sin x}$ $\frac{1}{\cos x} = \frac{3}{\sin x}$ $\frac{\sin x}{\cos x} = 3$ $\tan x = 3$ $x = 71,57^\circ + k \cdot 180; k \in \mathbb{Z}$	A✓ $\tan x = \frac{\sin x}{\cos x}$ A✓ $\sin^3 x = 3 \cos x - \sin x \cdot \cos^2 x$ CA✓ factors on LHS CA✓ equation CA✓ answer A✓ $+k \cdot 180; k \in \mathbb{Z}$ (6) OR A✓ $\tan x = \frac{\sin x}{\cos x}$ A✓ $\frac{\sin^2 x + \cos^2 x}{\cos x}$ CA✓ $\sin^2 x + \cos^2 x = 1$ CA✓ equation CA✓ answer A✓ $+k \cdot 180; k \in \mathbb{Z}$ (6)
		(6)
		[33]

QUESTION 6

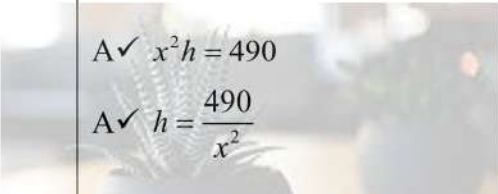
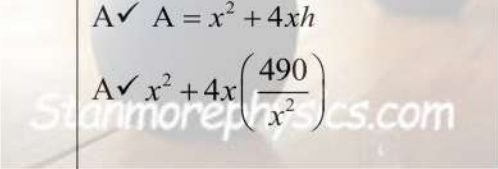
6.1	$a = 2$	A✓ answer (1)
6.2	Period = 180°	A✓ answer (1)
6.3		Graph of g: A✓ x-intercepts and y-intercept A✓ turning points A✓ shape
		(3)

6.4	$f(x) - 1 = g(x) + 2$ $f(x) = g(x) + 3$ $\therefore x = 0^\circ$	A✓ $f(x) = g(x) + 3$ A✓ answer (2)
6.5	$1 - 2[\sin(90^\circ - x)] = 1 - 2\cos x = -2\cos x + 1$ Minimum value of $-2\cos x$ is -2 Minimum value of $1 - 2\cos x = -2 + 1$ $= -1$ OR $1 - 2[\sin(90^\circ - x)] = 1 - 2\cos x$ Maximum value of $2\cos x$ is 2 $\therefore$ Minimum value of $1 - 2\cos x = -1$	A✓ $1 - 2\cos x$ A✓ min. of $-2\cos x$ A✓ answer (3) A✓ $1 - 2\cos x$ A✓ max of $2\cos x$ A✓ answer (3)
		[10]

QUESTION 7

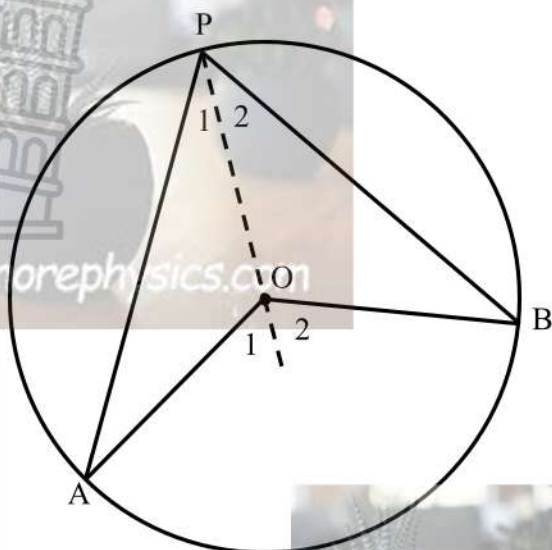
7.1	$\hat{RPS} = (\theta - \alpha)$	A✓ answer (1)
7.2	$\hat{S}_2 = (90^\circ + \alpha)$ [exterior angle of Δ] In ΔRPS : $\frac{RS}{\sin(\theta - \alpha)} = \frac{k}{\sin(90^\circ + \alpha)}$ $\therefore RS = \frac{k \cdot \sin(\theta - \alpha)}{\cos \alpha}$	A✓ $\hat{S}_2 = (90^\circ + \alpha)$ A✓ substitution into the sine rule A✓ $\sin(90^\circ + \alpha) = \cos \alpha$ (3)
7.3	$RS = \frac{17 \cdot \sin(58^\circ - 23^\circ)}{\cos 23^\circ}$ $= 10,59$ units	A✓ substitution CA✓ answer (2)
7.4	$\hat{R} = 90^\circ - 58^\circ$ [sum of $\angle$ s in a Δ] $= 32^\circ$ Area $\Delta RSP = \frac{1}{2}(RS)(PR)\sin \hat{R}$ $= \frac{1}{2} \times 10,59 \times 17 \times \sin 32^\circ$ $= 48$ square units	A✓ $\hat{R} = 32^\circ$ A✓ substitution into the area rule (2)
		[8]

QUESTION 8

8.1	$V = x^2 h$ $\therefore x^2 h = 490$ $h = \frac{490}{x^2}$	 A✓ $x^2 h = 490$ A✓ $h = \frac{490}{x^2}$	(2)
8.2	$A = x^2 + 4xh$ $= x^2 + 4x \left(\frac{490}{x^2} \right)$ $= x^2 + \frac{1960}{x} \text{ cm}^2$	 A✓ $A = x^2 + 4xh$ A✓ $x^2 + 4x \left(\frac{490}{x^2} \right)$	(2)
8.3	$\frac{490}{x^2} = 10$ $x^2 = 49$ $x = 7$ $A = (7)^2 + \frac{1960}{7}$ $= 329 \text{ cm}^2$	 A✓ substitution of $h = 10$ A✓ $x = 7$ CA✓ substitution into external surface area CA✓ answer	(4)
			[8]

QUESTION 9

9.1



RTP: $\hat{AOB} = 2\hat{P}$

Construction: Draw PO produced

Proof:

$$\hat{O}_1 = \hat{P}_1 + \hat{A}$$

$$\text{But } \hat{P}_1 = \hat{A}$$

$$\therefore \hat{O}_1 = 2\hat{P}_1$$

$$\text{Similarly: } \hat{O}_2 = 2\hat{P}_2$$

$$\therefore \hat{O}_1 + \hat{O}_2 = 2\hat{P}_1 + 2\hat{P}_2$$

$$= 2(\hat{P}_1 + \hat{P}_2)$$

$$\text{i.e. } \hat{AOB} = 2\hat{APB}$$

[ext. $\angle$ of Δ]

[$\angle$ s opp. = sides]

If there is no construction: $\frac{0}{5}$

✓ construction

✓ S/R

✓ S/R

✓ S

✓ S

(5)

OR

Construction: Draw PO

Proof:

$$\hat{A} = \hat{P}_1$$

[$\angle$ s opp. equal sides]

$$\therefore \hat{POA} = 180^\circ - 2\hat{P}_1$$

[sum of $\angle$ s of ΔAPO]

$$\hat{B} = \hat{P}_2$$

[$\angle$ s opp. equal sides]

$$\therefore \hat{POB} = 180^\circ - 2\hat{P}_2$$

[sum of $\angle$ s of ΔBPO]

$$\hat{AOB} = 360^\circ - (\hat{POA} + \hat{POB})$$

[$\angle$ s around a point]

$$= 360^\circ - (180^\circ - 2\hat{P}_1 + 180^\circ - 2\hat{P}_2)$$

$$= 2\hat{P}_1 + 2\hat{P}_2$$

$$= 2(\hat{P}_1 + \hat{P}_2)$$

$$= 2\hat{APB}$$

If there is no construction: $\frac{0}{5}$

✓ construction

✓ S/R

✓ S/R

✓ S

✓ S

(5)

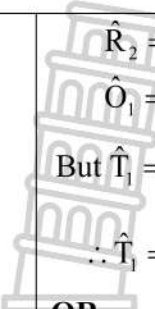
GRADE 11
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9.2.1	$\hat{R}_1 = \hat{S}_1 = 42^\circ$	[$\angle$ s opp. = sides]	✓ S ✓ R (2)
9.2.2 (a)	$\hat{PQR} = 96^\circ$	[opp. $\angle$ s of a cyclic quad]	✓ S ✓ R (2)
9.2.2 (b)	$\hat{SMR} = 96^\circ$ $\therefore \hat{P}_1 = 48^\circ$	[sum of angles of Δ] [$\angle$ at centre = $2 \times \angle$ at circumference]	✓ S/R ✓ S ✓ R (3)
9.2.3	$\hat{R}_3 = \hat{P}_1 = 48^\circ$ $\therefore SP \parallel QR$ OR $\hat{SRQ} = 96^\circ$ $\hat{PSR} + \hat{SRQ} = 84^\circ + 96^\circ$ $= 180^\circ$ $\therefore SP \parallel QR$	[equal chords; equal $\angle$ s] [alt $\angle$ s =] [opp. $\angle$ s of a cyclic quad supplementary] [co-int $\angle$ s supplementary]	✓ S ✓ R ✓ R (3) ✓ S ✓ R ✓ R (3) [15]

QUESTION 10

10.1	$\hat{OTV} = 90^\circ$ $\hat{OZV} = 90^\circ$ $\hat{OTV} + \hat{OZV} = 180^\circ$ $\therefore TOZV$ is a cyclic quadrilateral OR $TV = ZV$ $\therefore \hat{TZV} = \hat{T}_3 = x$ $\hat{OZV} = 90^\circ$ $\therefore \hat{Z}_3 = 90^\circ - x$ but $\hat{P}_2 = 90^\circ$ $\therefore \hat{O}_3 = x$ $\therefore \hat{O}_3 = \hat{T}_3$ $\therefore TOZV$ is a cyclic quadrilateral	[tan $\perp$ radius] [tan $\perp$ radius] [converse opp. $\angle$ s of a cyclic quad. suppl.] [two tans from same point] [$\angle$ s opp. = sides] [tan $\perp$ radius] [line from centre to midpoint] [sum of angles of Δ] [converse $\angle$ s in the same segment]	✓ S ✓ R ✓ S/R ✓ R (4) ✓ S/R ✓ S/R ✓ S/R ✓ R (4)
10.2	$\hat{RZT} = 90^\circ$ $\hat{P}_1 = 90^\circ$ $\therefore RZ \parallel SY$ OR $\hat{RZT} = 90^\circ$ $\hat{P}_2 = 90^\circ$ $\therefore RZ \parallel SY$	[$\angle$ in semi-circle] [line from centre to midpoint of chord] [corresponding $\angle$ s =] [angle in semi-circle] [line from centre to midpoint of chord] [co-interior $\angle$ s supplementary]	✓ S ✓ R ✓ S ✓ R ✓ R (5) ✓ S ✓ R ✓ S ✓ R ✓ R (5)

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10.3	 $\hat{R}_2 = x$ $\hat{O}_1 = \hat{R}_2 = x$ But $\hat{T}_1 = \frac{1}{2} \hat{O}_1$ $\therefore \hat{T}_1 = \frac{1}{2} x$ OR $\hat{R}_2 = x$ $\hat{O}_1 = \hat{R}_2 = x$ and $\hat{T}_1 = \hat{S}_1$ but $\hat{T}_1 + \hat{S}_1 = \hat{O}_1$ $\therefore \hat{T}_1 + \hat{T}_1 = x$ $\therefore 2\hat{T}_1 = x$ $\therefore \hat{T}_1 = \frac{1}{2} x$	[tan chord theorem] [alt $\angle$s = ; $RZ \parallel SY$] [$\angle$ at centre = $2 \times \angle$ at circumference]	✓ S ✓ R ✓ S/R ✓ S/R (4)
10.4	 $\hat{S}_1 = \hat{T}_1 = \frac{1}{2} x$ $\hat{Z}_2 = \hat{S}_1 = \frac{1}{2} x$ $\hat{Y}_2 = \hat{P}_3 + \hat{Z}_2$ $= 90^\circ + \frac{1}{2} x$ $= 90^\circ + \hat{T}_1$ $\hat{T}_1 = \frac{1}{2} x$ $\hat{T}_2 = 90^\circ - x$ but $\hat{Y}_1 = \hat{T}_1 + \hat{T}_2$ $\therefore \hat{Y}_1 = \frac{1}{2} x + 90^\circ - x$ $= 90^\circ - \frac{1}{2} x$ $\hat{Y}_2 = 180^\circ - \hat{Y}_1$ $= 180^\circ - \left(90^\circ - \frac{1}{2} x \right)$ $= 90^\circ + \frac{1}{2} x$ $= 90^\circ + \hat{T}_1$	 [$\angle$s opp. = sides] [$\angle$s in the same segment] [ext. $\angle$ of Δ] OR [proved in 10.3] [tan $\perp$ radius] [$\angle$s in the same segment] [$\angle$s on a straight line]	✓ S/R ✓ S ✓ R ✓ S/R ✓ S (5) OR ✓ S/R ✓ S ✓ R ✓ S/R ✓ S (5)

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	OR $\hat{T}\hat{S}R = 90^\circ$ $\therefore \hat{R}_1 = 180^\circ - 90^\circ - \frac{1}{2}x$ $\therefore \hat{R}_1 = 90^\circ - \frac{1}{2}x$ $\hat{Y}_2 = \hat{R}_1 + \hat{R}_2$ $= 90^\circ - \frac{1}{2}x + x$ $= 90^\circ + \frac{1}{2}x$ $= 90^\circ + \hat{T}_1$	OR [$\angle$ in a semi-circle] [sum of $\angle$s of a Δ] [ext. $\angle$ of a cyclic quad]	OR ✓ S ✓ R ✓ S/R ✓ S/R ✓ S
			(5)
			[15]

QUESTION 11

11.1	$OA = OB = k + m$ In $\triangle AOE$: $AE^2 = OA^2 - OE^2$ $= (k + m)^2 - k^2$ $= k^2 + 2km + m^2 - k^2$ $= 2km + m^2$ In $\triangle AEB$: $AB^2 = AE^2 + EB^2$ $= 2km + m^2 + m^2$ $= 2km + 2m^2$ OR In $\triangle AEB$: $AB^2 = EB^2 + AE^2$ In $\triangle OEA$: $AE^2 = OA^2 - OE^2$ $\therefore AB^2 = EB^2 + OA^2 - OE^2$ and $OA = OB$ $\therefore AB^2 = EB^2 + OB^2 - OE^2$ $= EB^2 + OB^2 - (OB - EB)^2$ $= EB^2 + OB^2 - OB^2 + 2OB \cdot EB - EB^2$ $= 2OB \cdot EB$ $= 2(k + m) \cdot m$ $= 2km + 2m^2$	[radii] [Pythagoras] [Pythagoras] [Pythagoras] [Pythagoras] [Pythagoras] [radii]	✓ S/R ✓ S/R ✓ S ✓ S ✓ S/R ✓ S/R ✓ S ✓ S
			(4)
			(4)
11.2	$DE = AE$ $= \sqrt{2km + 2m^2}$	[line from centre $\perp$ chord]	✓ R ✓ S
			(2)
			[6]

TOTAL: 150 MARKS